

Multi Objective Optimization in Recommendation systems

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Abstract

Recommender systems aim to provide personalized suggestions based on user preferences, enhancing user experiences across various domains, including e-commerce. In this study, we investigate the application of Multi-Objective Optimization (MOO) in recommendation systems, which seeks to optimize multiple objectives simultaneously [1]. Traditional recommender systems focus primarily on click-through rate (CTR), which measures the likelihood of user interaction with items. However, optimizing for CTR alone does not account for other important factors such as Gross Merchandise Volume (GMV), which measures revenue generation. These objectives often conflict, as improving one can degrade the other [2]. To address this issue, we explore three baseline approaches: the PE-LTR framework, Linear SVM, and RBF Kernel SVM [3]. We integrate these techniques to simultaneously optimize both CTR and GMV while maintaining a balance between these competing objectives. Our results demonstrate the comparative performance of these approaches through key metrics, such as Top-K accuracy, on several benchmark datasets. The findings suggest that the use of Multi-Objective Optimization with appropriate baselines can lead to more effective and balanced recommendations, improving both user engagement and revenue generation. This work contributes to the growing body of research on MOO in recommender systems and offers practical insights for e-commerce applications where multiple objectives must be optimized simultaneously.

1 Introduction

In modern e-commerce platforms, recommendation systems serve as crucial tools for enhancing user experience and driving business value [8, 9]. These systems face the complex challenge of generating personalized item rankings that must simultaneously satisfy multiple, often competing, objectives [10]. The fundamental challenge lies in developing algorithms that can effectively balance these objectives while maintaining high-quality recommendations.

1.1 Problem Definition and Motivation

Recommendation algorithms are tasked with generating personalized item rankings, where the highest-ranked items are presented to users. The core challenge emerges from the inherent tension between user interests and business metrics [11]. In e-commerce recommendation systems, two critical metrics stand out: Click-Through Rate (CTR) and Gross Merchandise Volume (GMV) [12]. CTR represents the proportion of users who interact with a recommended item relative to the total number of users who view the recommendation. This metric serves as a primary indicator of recommendation relevance and user engagement. GMV, conversely, quantifies the total value of merchandise sold, directly reflecting revenue generation. These metrics often exhibit an inverse relationship, creating a complex optimization challenge [13].

The conflict between these objectives manifests in several ways. When systems optimize solely for CTR, they tend to recommend lower-priced items that users are more likely to click on, potentially leading to reduced GMV. Empirical analysis of our dataset reveals a statistically significant negative correlation (correlation coefficient -0.407, p-value <0.01) between CTR

and GMV when optimizing for CTR alone, as illustrated in Figure 1.

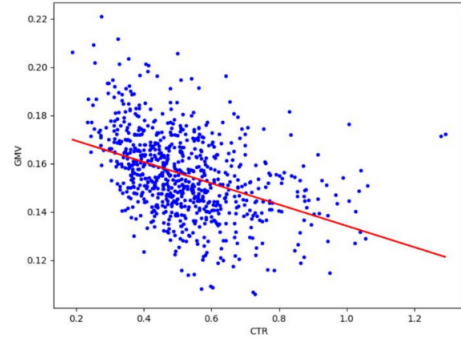


Figure 1. Correlation between the CTR and GMV

1.2 Multi-Objective Optimization Framework

To address these competing objectives, we formulate the problem within a multi-objective optimization framework. Formally, a multi-objective optimization problem can be expressed as:

$$\min_{\substack{g_j(x) \geq 0 \forall j=1,2,\dots,J \\ h_i(x) = 0 \forall i=1,2,\dots,L}} (\Phi_1(x), \Phi_2(x), \dots, \Phi_M(x)) \quad (1)$$

where $(\Phi_1(x), \Phi_2(x), \dots, \Phi_M(x))$ represent M distinct objectives. A solution is considered optimal when no objective can be improved without degrading at least one other objective.

1.3 Proposed Approaches

In this work, we explore three distinct approaches to address the multi-objective optimization challenge:

1.3.1 PE-LTR Framework

The PE-LTR framework offers a novel approach to balancing multiple objectives through weighted optimization. It employs a scalarization technique that combines multiple objectives into a single scalar objective function:

$$\Phi(x) = \alpha_1 \Phi_1(x) + \alpha_2 \Phi_2(x) + \dots + \alpha_M \Phi_M(x) \quad (2)$$

where $\alpha_1, \alpha_2, \dots, \alpha_M$ represent the weights assigned to each objective.

1.3.2 Support Vector Machine Approaches

We implement and evaluate two SVM-based approaches: 1. Linear SVM: This approach provides a baseline implementation that captures linear relationships between features while maintaining computational efficiency. 2. RBF Kernel SVM: By employing a Radial Basis Function kernel, this method captures non-linear relationships between features, potentially offering improved performance for complex recommendation patterns.

Our experimental evaluation demonstrates the effectiveness of these approaches across multiple metrics, including Top-K accuracy, with systematic comparison across various baseline implementations. The results indicate that while each approach offers distinct advantages, the choice of method depends significantly on the specific requirements and constraints of the recommendation system.

Through extensive experimentation on real-world datasets, we demonstrate that these approaches can effectively balance the competing objectives of CTR and GMV, leading to more robust and practical recommendation systems for e-commerce applications.

2 Preliminaries

In multi-objective optimization, we seek to optimize multiple conflicting objective functions simultaneously. Let $\Phi_1(x), \Phi_2(x), \dots, \Phi_M(x)$ represent the M objective functions. A solution x is said to be dominated by another solution x^* if the following conditions hold:

1. For all objectives $m = 1, 2, \dots, M$:

$$\Phi_m(x^*) \leq \Phi_m(x)$$

2. There exists at least one objective m' such that:

$$\Phi_{m'}(x^*) < \Phi_{m'}(x)$$

The first condition ensures that all objectives at x^* are no worse than those at x , while the second condition guarantees that x^* is strictly better in at least one objective. This relationship is visually represented in Figure 2, where solution A dominates solutions C and D, and solution B dominates solution C. However, there is no dominance relationship between solutions A and B or between D and B.

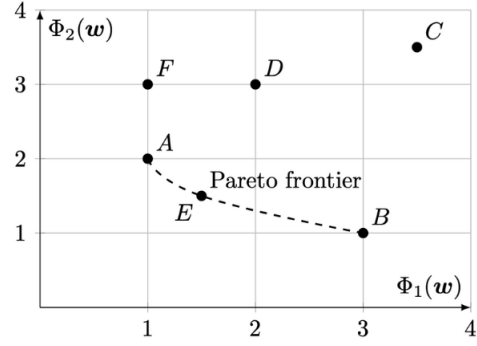


Figure 2. illustration of the Pareto dominance relation

A solution that is not dominated by any other solution is termed **Pareto optimal** or **Pareto efficient**. In the context of multi-objective optimization, optimality is defined through Pareto optimality. In Figure 2, solutions A, E, and B are identified as Pareto optimal solutions.

The **Pareto front** or **Pareto frontier** is constructed by connecting all Pareto optimal solutions with a curve. In this example, joining points A, E, and B forms the Pareto frontier.

The **Pareto set** refers to the collection of all Pareto optimal solutions. Furthermore, a point $\hat{x}$ is classified as **Pareto stationary** if it satisfies the condition:

$$\sum_{i=1}^M \alpha_i \nabla_x \Phi_i(\hat{x}) = 0,$$

where $\alpha_i \geq 0$ and $\sum_{i=1}^M \alpha_i = 1$.

It is important to note that if a point x is Pareto optimal, then it must also be Pareto stationary; however, the converse does not hold true:

$$\text{Pareto optimal} \Rightarrow \text{Pareto stationary},$$

but

$$\text{Pareto stationary} \not\Rightarrow \text{Pareto optimal}.$$

This framework provides a foundational understanding of how solutions can be evaluated and compared in multi-objective optimization scenarios, particularly within recommendation systems where trade-offs between objectives are prevalent.

3 Related Work

3.1 Model Distillation for Multi-Objective Optimization in Recommendation Systems

The paper[4] proposes a model distillation-based approach to address challenges in multi-objective optimization for recommendation systems, particularly in ranking problems. It focuses on balancing primary objectives (e.g., conversion rates) with secondary objectives (e.g., cancellations, long-term growth) in a unified framework.

3.1.1 Problem Formulation

Given a search query q_i with n matched items, the task is to assign scores s_j to items $t_{i,j}$ such that they can be ranked in descending order. The training dataset includes feature vectors $X_i = \{x_{i,j}\}_{j=1}^n$ and labels $L_i = \{l_{i,j}\}_{j=1}^n$, where each label corresponds to an objective.

The cost function for each objective is defined as:

$$C_k(X) = \sum_i \text{loss}(f(\theta, X_i), L_{i,k}),$$

where $f(\theta, X_i)$ represents model predictions and $L_{i,k}$ represents labels for the k -th objective.

3.1.2 Model Distillation Framework

The framework extends multi-objective optimization (MOO) into a model distillation paradigm, leveraging the similarity between distillation and scalarization methods. The primary objective is optimized while ensuring secondary objectives are satisfied as constraints:

$$\min_{\theta} C_1(f(\theta, X))$$

$$\text{subject to } |f(\theta, X) - f_k(\theta_k^*, X)| \leq \epsilon_k, \forall k > 1,$$

where $f_k(\theta_k^*, X)$ denotes the optimal model for objective k . Using Lagrangian relaxation, this is reformulated as:

$$\min_{\theta} C_1(f(\theta, X)) + \sum_{k=2}^K \omega_k |f(\theta, X) - f_k(\theta_k^*, X)|,$$

where ω_k are weights controlling the trade-offs between objectives.

To simplify training, the authors introduce *soft labels*:

$$\text{Soft-label: } \sum_{k=1}^K \omega_k f_k(\theta_k^*, X),$$

which aggregates predictions from pre-trained models for each objective.

3.1.3 Key Contributions

- **Soft Labels for Multi-Objective Learning:**
 - By replacing hard constraints with soft labels, the system ensures smooth optimization for both primary and secondary objectives.
 - The soft label acts as a regularizer, mitigating issues of imbalanced datasets and irreproducibility.
- **Integration of Ad-hoc Business Objectives:**
 - The framework accommodates non-differentiable objectives (e.g., promoting new items) by revising soft labels rather than modifying the model directly.
- **Self-Distillation:**
 - Soft labels are reused across iterations to transfer knowledge from previous models, reducing the need for retraining pre-trained models.

3.1.4 Significance in Recommendation Systems

This framework demonstrates how model distillation can streamline multi-objective optimization in recommendation systems. It balances competing objectives, accommodates ad-hoc business rules, and enhances scalability through efficient knowledge transfer.

3.2 Whole Page Optimization with Global Constraints

Whole Page Optimization (WPO)[5] addresses the challenge of ranking and organizing widgets on web homepages to balance multiple competing objectives. In this framework, the Amazon Video homepage serves as a test case, requiring optimization of three dimensions: **relevance**, **diversity**, and **business constraints**.

3.2.1 Problem Formulation

The authors formalize the WPO problem using a *joint optimization framework* that maximizes utility across all homepage contexts while ensuring global constraints. Let:

- C_u^n : Top n widgets selected for a given context u .
- $f(C_u^n|u, w)$: Utility or scoring function based on user context u and model parameters w .
- $g(C_u^n)$: Constraint function applied across all homepages.

The optimization is framed as:

$$\max_{C_u^n: u \in U_t} \sum_{u \in U_t} f(C_u^n|u, w)$$

subject to:

$$\sum_{u \in U_t} g(C_u^n) \leq 0,$$

where U_t represents all homepage contexts during time period t .

This formulation ensures maximization of utility while satisfying business constraints.

3.2.2 Key Contributions

The paper introduces a *primal-dual algorithm* to decouple offline optimization (learning global constraints) from online page composition. The scoring function $f(C_n|u, w)$ incorporates **submodular diversity features**, which promote widget diversity by diminishing marginal gains for repeated categories.

Submodular Diversity:

$$\rho(x) = \log(1 + x),$$

where $\rho(x)$ diminishes incremental utility for repeated widget categories.

Primal-Dual Formulation: The dual problem is expressed as:

$$\max_{\xi \geq 0} \min_{C_u^n} - \sum_{u \in U_t} \left[\sum_{i=1}^n f(c_i|C_{i-1}, u, w) + \xi^\top (b - \sum_{i=1}^n I(c_i)) \right],$$

where ξ are dual variables regulating trade-offs between utility and constraints.

3.2.3 Significance in Recommendation Systems

The WPO framework provides a scalable approach to recommendation systems that require balancing relevance and diversity while meeting business goals. Its incorporation of **submodular diversity** and **global constraint satisfaction** sets a precedent for optimizing multi-faceted recommendation tasks.

3.3 Pareto-Efficient Learning-to-Rank for Multi-Objective Optimization

The challenge of balancing conflicting objectives in recommendation systems has led to the development of Pareto-efficient frameworks, where the goal is to optimize multiple objectives without degrading any individual metric. This paper introduces a novel algorithmic framework, **Pareto-Efficient Learning-to-Rank (PE-LTR)**, designed to achieve Pareto-optimal solutions for multi-objective recommendation problems [6].

PE-LTR leverages the scalarization technique to aggregate multiple differentiable loss functions into a single objective. To ensure Pareto efficiency, the framework incorporates a theoretically grounded condition for the scalarization weights, transforming the problem into a constrained quadratic programming task. A two-step algorithm is employed to solve this, first deriving an approximate solution through relaxation and then refining it using a projection step to meet feasibility constraints.

The framework demonstrates its scalability and flexibility by accommodating various models and objectives, as long as gradients are available. In application to e-commerce recommendation, PE-LTR optimizes **Gross Merchandise Volume (GMV)** and **Click-Through Rate (CTR)**—two critical yet often conflicting objectives. The system uses differentiable formulations for GMV and CTR, enabling simultaneous optimization through a unified model. Unlike conventional methods that rely on separate models for different objectives, PE-LTR ensures consistency by learning both objectives within a single framework.

The framework also facilitates the generation of Pareto Frontiers, enabling stakeholders to select trade-off solutions tailored to their specific priorities, such as fairness metrics. Extensive offline and online experiments validate the superiority of PE-LTR, showing significant improvements in GMV metrics while maintaining competitive CTR performance. Compared to state-of-the-art methods like LambdaMART and PO-EA, PE-LTR achieves higher G-NDCG and G-MAP scores, underscoring its effectiveness in managing multi-objective trade-offs.

This work establishes PE-LTR as a robust and scalable solution for real-world multi-objective optimization challenges in recommendation systems, with its theoretical guarantees and adaptability making it a

valuable tool for e-commerce platforms.

4 Methodology

4.1 Dataset Description

4.1.1 Dataset Overview

The dataset used for this study is derived from a product ranking task on an e-commerce platform, where products are ranked on a product page based on user interaction. It contains **373,473 unique requestIds**, which represent session IDs of user visits. Each **requestId** contains multiple **impressionIds**, representing the product IDs associated with that request.

From the analysis of the dataset:

- Approximately **353,671 requestIds** have **exactly one isConversion=1**, indicating successful conversions (i.e., purchases).
- About **266,179 requestIds** have **exactly one isClick=1**, indicating user engagement with products (i.e., clicks).
- It is evident from the dataset that **isClick=1** always occurs whenever **isConversion=1**. This signifies that a purchase is always preceded by a click.
- However, there are around **283,237 requestIds** where there are **no instances of isClick=1 and isConversion=0**, i.e., sessions where a user clicked on a product but did not make a purchase are relatively rare. This observation suggests a high engagement-to-conversion ratio in this dataset.

4.1.2 Feature Definitions

The dataset includes several key features that facilitate ranking products on a product page. The features are defined as follows:

- **requestId**: Session ID of a visit, treated as a single visit for a user. A single user may have multiple **requestIds** across different sessions.
- **impressionId**: Product IDs within a request, representing products displayed to the user.
- **pCTR**: Predicted Click-Through Rate (CTR) from the CTR model, representing the probability of a user clicking on a product.
- **pCVR**: Predicted Conversion Rate (CVR) from the CVR model, representing the probability of a user purchasing a product after clicking on it.
- **asp**: The price of the product being displayed.
- **isClick**: Binary label indicating whether a product was clicked by the user.
- **isConversion**: Binary label indicating whether a product was purchased by the user.

4.1.3 Baseline Score Formulation

The dataset is accompanied by a baseline ranking formulation for calculating the final score of each product. This score is derived from the predicted outputs of the CTR and CVR models, as well as the product price (**asp**). The baseline formulation is expressed as:

$$\begin{aligned} \text{final_score} = & 4.370724896 \cdot \text{pCTR} \\ & + 739.8453433 \cdot \text{pCTR} \cdot \text{pCVR} \\ & + 0 \cdot \text{pCTR} \cdot \text{pCVR} \cdot \text{asp} \end{aligned}$$

- **pCTR:** The weight associated with predicted click probabilities contributes directly to the ranking score.
- **pCTR · pCVR:** The product of click-through and conversion probabilities, scaled by a significant factor, emphasizes the likelihood of a user progressing from a click to a purchase.
- **pCTR · pCVR · asp:** Although included in the formulation, the term involving product price is assigned a weight of 0 in the baseline, implying that price is not factored into the score in this initial formulation.

This baseline formulation is used to rank products within a request and provides a foundation for evaluating the ranking methodology and optimizing it further for improved performance.

4.1.4 Analysis of Clicks and Conversions

To gain deeper insights into user interactions, we analyzed the distribution of clicks and conversions per `requestId`. The results are summarized in the following tables.

Table 1: Distribution of Clicks per `requestId`

Number of Clicks per <code>requestId</code>	Count of <code>requestId</code>
1	266,179
2	73,805
3	23,283
4	7,371
5	2,164
6	518
7	108
8	38
9	4
10	2
14	1

Table 2: Distribution of Conversions per `requestId`

Number of Conversions per <code>requestId</code>	Count of <code>requestId</code>
1	353,671
2	15,456
3	3,142
4	867
5	260
6	50
7	10
8	13
9	1
10	1
11	1
14	1

Table 3: Distribution of Clicks Without Conversion per `requestId`

Number of Clicks (No Conversion)	Count of <code>requestId</code>
0	283,237
1	62,654
2	19,337
3	6,029
4	1,719
5	401
6	76
7	20

The tables illustrate the following:

- **Table 1** shows that the majority of `requestIds` have a small number of clicks, with over 266,000 sessions having exactly one click.
- **Table 2** indicates a high number of `requestIds` with at least one conversion, highlighting strong user engagement.
- **Table 3** reveals that a significant number of sessions (over 283,000) have zero instances where a product was clicked but not purchased, suggesting that non-converting clicks are relatively rare.

These observations support the conclusion that the dataset exhibits a high engagement-to-conversion ratio, with most user sessions resulting in purchases after clicks.

4.2 Multi-objective Optimization using PE-LTR on Recommendation Dataset

This subsection delves into the application of the Pareto-Efficient Learning-to-Rank (PE-LTR) framework for multi-objective optimization in recommendation systems. The objectives include optimizing the `isClick` and `isConversion`.

4.2.1 Objective Functions

Logistic Loss: The optimization is driven by logistic loss functions for both objectives. The logistic loss for an objective y is given by:

$$L(y, \hat{y}) = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

where $\hat{y}_i = \frac{1}{1 + \exp(-z_i)}$ and $z_i = X \cdot \theta$.

For `isClick` and `isConversion`:

$$\begin{aligned} L_{\text{isClick}}(\theta) = & -\frac{1}{N} \sum_{i=1}^N w_{\text{click},i} \left[y_{\text{click},i} \log(\hat{y}_{\text{click},i}) \right. \\ & \left. + (1 - y_{\text{click},i}) \log(1 - \hat{y}_{\text{click},i}) \right] \end{aligned}$$

$$\begin{aligned} L_{\text{isConversion}}(\theta) = & -\frac{1}{N} \sum_{i=1}^N w_{\text{conversion},i} \left[y_{\text{conversion},i} \log(\hat{y}_{\text{conversion},i}) \right. \\ & \left. + (1 - y_{\text{conversion},i}) \log(1 - \hat{y}_{\text{conversion},i}) \right] \end{aligned}$$

Weights w_{click} and $w_{\text{conversion}}$ are used to handle class imbalance and assign higher importance to positive samples.

Scalarized Loss: Using scalarization, the combined objective is formulated as:

$$L_{\text{combined}}(\theta, \omega) = \omega_{\text{isClick}} L_{\text{isClick}}(\theta) + \omega_{\text{isConv}} L_{\text{isConv}}(\theta)$$

where ω_{isClick} and $\omega_{\text{isConversion}}$ are weights satisfying $\omega_{\text{isClick}} + \omega_{\text{isConversion}} = 1$.

4.2.2 PE-LTR Algorithm

Pareto-Efficient Condition: The weights ω are optimized using the Karush-Kuhn-Tucker (KKT) conditions to ensure Pareto efficiency:

$$\min_{\omega} \left\| \sum_{i=1}^K \omega_i \nabla_{\theta} L_i(\theta) \right\|_2^2 \quad \text{s.t.} \quad \sum_{i=1}^K \omega_i = 1, \omega_i \geq c_i$$

The above is solved using a two-step procedure:

- Relax the constraints to find an analytical solution.
- Apply projection to ensure non-negativity and sum constraints.

Gradient Descent: The model parameters θ are updated using the Adam optimization rule:

$$\theta_{t+1} = \theta_t - \eta \frac{m_t}{\sqrt{v_t} + \epsilon}$$

where m_t and v_t are moving averages of the gradient and its square.

Feature Engineering: Additional features were derived to improve the model's predictive capabilities:

- Interaction terms: $\text{pCTR}_i \cdot \text{pCVR}_i$
- Quadratic terms: $\text{pCTR}_i^2, \text{pCVR}_i^2$

Manual Feature Standardization: Each feature was normalized manually to ensure numerical stability during optimization. For a feature x , standardization was applied as:

$$x_{\text{standardized}} = \frac{x - \mu}{\sigma}$$

where μ is the mean and σ is the standard deviation of the feature across all samples.

4.2.3 Evaluation Metric: Top-K Accuracy

The Top-K metric evaluates the performance of the ranking system by measuring the proportion of instances where a relevant item (i.e., $y_{\text{conversion}} = 1$) is found in the top K predictions for a request. For a given request i , the Top-K accuracy is calculated as:

$$\text{Top-K Accuracy} = \frac{\text{Number of requests with at least one relevant item in Top-K}}{\text{Total number of requests}} \times 100$$

This metric ensures that the ranking system prioritizes conversions in the top-ranked items for each request.

4.2.4 Algorithm: Pareto-Efficient Learning-to-Rank

Input: Features X , labels $y_{\text{click}}, y_{\text{conversion}}$, initial θ, ω

Output: Optimized θ , Pareto-efficient weights ω

1. Initialize θ, ω uniformly.
2. For each epoch:
 - (a) Compute $L_{\text{isClick}}, L_{\text{isConversion}}$.
 - (b) Compute gradients $\nabla_{\theta} L_{\text{isClick}}, \nabla_{\theta} L_{\text{isConversion}}$.
 - (c) Update θ using the gradient descent rule.
 - (d) Solve for ω using the Pareto-Efficient Condition (PEC) solver.
 - (e) Rank products within each request using $X \cdot \theta$.
 - (f) Calculate the Top-K metric for validation.
3. Return θ, ω .

4.2.5 PEC Solver

The Pareto-Efficient Condition (PEC) solver is a key component in ensuring Pareto efficiency. It adjusts the weight vector ω dynamically to balance the trade-off between isClick and isConversion optimization. The process involves:

- **Gradient Matrix Construction:** Gradients for isClick and isConversion losses are combined into a gradient matrix G :

$$G = [\nabla_{\theta} L_{\text{isClick}} \quad \nabla_{\theta} L_{\text{isConversion}}]$$

- **Optimization Objective:** The goal is to minimize the magnitude of the combined gradient vector $G \cdot \omega$:

$$\min_{\omega} \|G \cdot \omega\|_2^2$$

- **Constraints:** The optimization is performed under the following constraints:
 - $\sum_{i=1}^K \omega_i = 1$: Sum-to-one constraint
 - $\omega_i \geq c_i$: Non-negativity constraint with minimum weight
- **Quadratic Programming Solution:** The above problem is solved using quadratic programming methods to determine the optimal weight vector ω . This ensures a balanced contribution from both objectives while adhering to Pareto efficiency.

4.3 Support Vector Machines for Recommendation

4.3.1 Problem Reformulation

Due to the data distribution, where cases with $\text{isClick} = 1$ and $\text{isConversion} = 0$ are rare, we reformulated the problem as a binary classification task:

- **Positive Class:** Instances where $\text{isConversion} = 1$ (indicating successful conversions).
- **Negative Class:** Instances where $\text{isConversion} = 0$ (indicating no conversions).

This reformulation simplifies the classification task while aligning with the system's goal of identifying successful conversions.

4.3.2 Linear SVM

The Linear SVM aims to find a hyperplane that maximizes the margin between the positive and negative classes. The optimization problem is formulated as:

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^N \max(0, 1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b)),$$

where:

- $\mathbf{x}_i \in \mathbb{R}^d$: Input features.
- $y_i \in \{-1, +1\}$: Binary labels for the negative and positive classes.
- $\mathbf{w}$: Weight vector defining the hyperplane.
- b : Bias term.
- C : Regularization parameter balancing margin maximization and error minimization.

The decision function is:

$$f(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b,$$

and the predicted class is determined by the sign of $f(\mathbf{x})$. For ranking, the decision score $f(\mathbf{x})$ is used to order instances within each `requestId` group.

4.3.3 RBF Kernel SVM

To capture non-linear patterns, the RBF Kernel SVM maps input features into a higher-dimensional space using the RBF kernel:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2),$$

where $\gamma > 0$ is the kernel parameter controlling the kernel's spread.

The optimization problem in the dual form is:

$$\min_{\alpha} \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j y_i y_j K(\mathbf{x}_i, \mathbf{x}_j) - \sum_{i=1}^N \alpha_i,$$

subject to:

$$0 \leq \alpha_i \leq C, \quad \sum_{i=1}^N \alpha_i y_i = 0,$$

where α_i are dual variables. The decision function is:

$$f(\mathbf{x}) = \sum_{i=1}^N \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}) + b.$$

To handle large datasets, the RBF kernel was approximated using explicit feature mapping into a higher-dimensional space, enabling a linear classifier to operate efficiently in the transformed space.

4.3.4 Evaluation Metric: Top-K Accuracy

The effectiveness of the models was evaluated using **Top-K Accuracy**, which measures the ability to rank instances such that at least one positive class instance appears in the top-K predictions for each `requestId` group. The metric is computed as:

1. Group instances by `requestId`.

2. Rank instances within each group based on the decision scores $f(\mathbf{x})$ in descending order.
3. Count groups where at least one positive instance ($y = 1$) is present in the top-K predictions.
4. Compute the percentage of such groups over the total number of groups.

Mathematically, for a given K :

$$\text{Top-K Accuracy} = \frac{1}{|\mathcal{R}|} \sum_{r \in \mathcal{R}} \mathbb{I} \left(\max_{i \in \text{Top-K}(r)} y_i = 1 \right),$$

where $\mathcal{R}$ is the set of all `requestIds`, and $\mathbb{I}$ is an indicator function equal to 1 if the condition is true.

This metric evaluates how effectively the models rank positive class instances at the top, critical for recommendation systems.

5 Results

5.1 Performance Evaluation

The performance of the methods was evaluated using the **Top-K Accuracy** metric, where K represents the number of top-ranked items considered for evaluation. The results for $K = 1, 2, 5$ are summarized below, showcasing the mean and standard deviation values across the 10 test datasets:

Top-1 Accuracy

- Baseline Method: 20.6635 ± 0.1142
- PE-LTR: 20.9820 ± 1.4410
- Linear SVM: 21.3447 ± 0.3159
- RBF Kernel SVM: 22.3926 ± 0.1878

Top-2 Accuracy

- Baseline Method: 31.8671 ± 0.1224
- PE-LTR: 36.5761 ± 2.1168
- Linear SVM: 37.5776 ± 0.3039
- RBF Kernel SVM: 38.7595 ± 0.2128

Top-5 Accuracy

- Baseline Method: 85.7016 ± 0.0615
- PE-LTR: 88.5759 ± 0.4844
- Linear SVM: 89.1493 ± 0.0846
- RBF Kernel SVM: 89.3386 ± 0.0910

5.2 Observations

- The **RBF Kernel SVM** consistently achieved the highest Top-K Accuracy across all metrics, demonstrating its capability to model complex, non-linear patterns in the dataset.
- The **Linear SVM** also performed competitively, especially in comparison to the PE-LTR framework, which suggests that even simple decision boundaries can achieve robust performance in linearly separable datasets.
- The **Baseline Method** served as a reference, providing lower Top-K Accuracy compared to the other approaches. This highlights the benefits of adopting more advanced optimization techniques like SVMs.
- Across all methods, the standard deviations were minimal, particularly for RBF Kernel SVM

and Linear SVM, indicating consistent performance across the 10 test datasets.

Conclusion

This study explored the application of Multi-Objective Optimization (MOO) techniques to recommendation systems, focusing on balancing the *isClick* and *isConversion* objectives. By employing frameworks such as Pareto-Efficient Learning-to-Rank (PE-LTR) and support vector machines (Linear and RBF Kernel), we demonstrated their capability to address the inherent trade-offs between user engagement (*isClick*) and successful conversions (*isConversion*). Experimental results highlighted that the RBF Kernel SVM consistently achieved the highest Top-K accuracy, indicating its strength in modeling complex, non-linear interactions. These findings emphasize the importance of advanced optimization methods in creating effective and balanced recommendation systems.

Future Work

Future research could build upon this work by exploring the following directions:

- **Dynamic Objective Balancing:** Implementing mechanisms to dynamically adjust the balance between the *isClick* and *isConversion* objectives in real-time, based on user behavior and business needs.
- **Personalization of Objectives:** Tailoring optimization priorities for specific user groups or product categories, allowing a more granular trade-off between engagement and conversion metrics.
- **Integration of Additional Objectives:** Including other objectives such as session duration or repeat purchases alongside *isClick* and *isConversion* to further refine the recommendation quality.
- **Advanced Feature Engineering:** Incorporating deeper feature interactions and contextual signals to improve the predictive capability of both *isClick* and *isConversion*.
- **Application in Dynamic Environments:** Testing the proposed methods in domains with rapidly evolving preferences.

These advancements will provide more robust and adaptable solutions for multi-objective recommendation systems, enabling greater alignment between user engagement and business outcomes.

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